

Pointers

Lab questions, set 3

21. Write a C program to traverse and print all the elements of the following 2D array using a pointer to an array.

```
int a[2][3] = {  
    {10, 20, 30},  
    {40, 50, 60}  
};
```

22. Write a C program to calculate and display the sum of all elements of the following 2D array using pointer arithmetic

```
int a[3][2] = { {1, 2},  
                {3, 4},  
                {5, 6} };
```

23. Write a C program to find and display the row-wise sum of the following 2D array using pointers.

```
int a[3][3] = {  
    {2, 4, 6},  
    {1, 3, 5},  
    {7, 8, 9}  
};
```

24. Using pointer notation, write a C program to display the column-wise elements of the following 2D array.

```
int a[2][4] = {  
    {5, 10, 15, 20},  
    {25, 30, 35, 40}  
};
```

25. Write a C program to find the maximum element in the following 2D array using pointer traversal.

```
int a[3][4] = {  
    {12, 45, 23, 9},  
    {34, 56, 78, 11},  
    {10, 8, 6, 90}  
};
```

26. Write a C program to print the diagonal elements of the following square matrix using pointers.

```
int a[3][3] = {  
    {1, 2, 3},  
    {4, 5, 6},  
    {7, 8, 9}  
};
```

27. Write a C program to count the number of even elements in the following 2D array using pointer arithmetic.

```
int a[2][3] = {  
    {4, 7, 10},  
    {13, 16, 19}  
};
```

28. Using pointers, write a C program to print the transpose of the following matrix. Do the in-place transpose (don't use an extra array).

```
int a[3][3] = {  
    {1, 2, 3},  
    {4, 5, 6},  
    {6, 7, 8}};
```

29. Write a C program to copy the contents of the following 2D array into another 2D array using pointers.

```
int a[2][2] = {  
    {5, 10},  
    {15, 20}  
};
```

30. Write a C program to traverse the following 2D array using a single pointer (int *) and display the elements row-wise.

```
int a[3][2] = {  
    {11, 22},  
    {33, 44},  
    {55, 66}  
};
```